
THREE BITS WALKED: THE COMPLETED FORMAL CLOSURE OF THE SEVEN MILLENNIUM ROWS

*The Root Proved Unconditionally on No Axiom and No Posit; Block, Hypothesis and Resolution
Recorded Row by Row; Each Row Closed to One Act*

FOUNDATIONS OF MATHEMATICS · SUPPLY THEORY · MINI PAPER

One bit, one bit, one bit

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ABSTRACT. The root every row shares, to exist is to actuate, is proved in Appendix R unconditionally, at theorem grade, on no axiom and no posit: a theorem with no hypothesis on its constructed domain, and one closed law for every root that grounds itself. Each row's closing theorem consumes one reading of that root and no other posit; on the Riemann row the functional equation is a field of the closing theorem's input type (`ActualZeros.fold_closed`), cited, not posited; on the Birch and Swinnerton-Dyer row the reflection law of the centred expansions enters the sign theorem's frame (`SignedFrame`, consumed by `sign_fixes_curve_parity`), cited, and neither input type of that row's closure, `ActualCurves` or `ActualCurvesFull`, carries a cited theorem as a field. The root alone decides no value, by theorem; the root read on each row is that row's value, by theorem; and the derivation of each value from the register alone is blocked, by theorem, each in the row's own kernel. The Offering Bit (Islam 2026a) stated a three-bit arc for every blocked problem row: a block, that an even register cannot decide an odd target; a hypothesis, that exactly one orientation bit is missing at a located seat; and a resolution, that the bit enters by deed. One Bit Across the Wall (Islam 2026b) typed twenty-three rows under that arc, and Nothing Escapes Least Erasure (Islam 2026j) graded them on two channels, kinetic and formal. This mini paper is the upgrade that reports the closure papers. By 5 October 2026 the author had published the first editions of seven closure papers, one per Millennium row; the editions of record, dated 6 October 2026, are the ones reported here, each with a Lean 4 kernel whose closing theorem, the one that prints the row's value from its act, rests on no axiom. We record them as the arc's results and grade them on the two channels. The block bit is proved on all seven rows, each paper formalizing its row's traditional barrier. The hypothesis bit is settled on all seven: on the Riemann, Hodge, Poincaré and Birch and Swinnerton-Dyer rows the seat is located in the kernel with its one missing orientation, and its parity where the row has one; on Navier–Stokes and Yang–Mills no seat is formalized, and the one bit, on Yang–Mills one per part, is carried through the record wall; on P versus NP no orientation sits at a seat: the complementation seat is proved vacant, and the hypothesis bit is the keyed separation bit, which no reading of the vanished seat's record returns. The resolution bit is supplied on all seven by the act, existence read on the row, proved equal to the row's value, and on each row its closure paper leaves nothing remaining on its closure. The kinetic channel is crossed by the act at premise grade, on the root proved beneath it, on every row but Poincaré, where that premise is discharged by Perelman's witness at the grade of its citation, a theorem of the literature carried as the witness, in the shape the kernel types (`Witness`, consumed by `witness_gives_act`), the kernel constructing no witness, so that the row stands at the grade of that citation (crossed; discharged, at the grade of its citation); the formal channel, the traditional route of derivation from the register's resources, is blocked by theorem on every open row, and Poincaré alone is crossed on it, by Perelman's offering, at the grade of its citation (Perelman 2002), full grade in the carried ledger of Islam (2026j). One cumulative ledger, the twenty-three rows of the earlier volume with the closure papers placed on top, each closure paper governing its row over the carried ledger whatever the edition dates, records every row's block, hypothesis and resolution bits, channels, remainder and physical witness; the target bit and the even register stand in Islam (2026b, Table 2), cited, not reprinted.

KEYWORDS *odd-supply separation · one-cut hypothesis · offering bit · Millennium Prize problems · existence · two channels · Lean 4*

The root every row of this paper shares is proved, on no axiom and no hypothesis, in Appendix R. Each row's closing theorem consumes one reading of that root, on its row, and no other posit, the Riemann row's input type

also carrying the functional equation, cited, and its axiom cone is empty. The root alone decides no value, by theorem; the root read on the row is the value, by theorem; and the traditional route of derivation from the register

is blocked, by theorem.

1. INTRODUCTION

The Offering Bit (Islam 2026a) closed its arc in three bits, each irreducible. The first is a theorem: on a finite frame with a native involution, a reading register that is even under the involution cannot decide an odd target, and the undecided remainder at the target's seat is exactly one binary orientation (Islam 2026a, Theorem 4.1 and Corollary 4.4). The second is a hypothesis: on every blocked row the missing item is one provenance-clean orientation supply, dated at entry (Hypothesis 11.1). The third is the resolution: that the bit enters by deed. Its theory, a dated-entry recursion that halts at the unique finest partition, typing every odd target at the halt as crossed, restatement-blocked, or defeater-absent, is a theorem (Part Three, Theorem 4.4); the bit itself is supplied, never derived. No bit can be skipped and none can be split. The count is fixed by the parity law, the one-bit corollary and the crossing theorem, and every historical crossing passed through all three.

One Bit Across the Wall (Islam 2026b) carried the arc to twenty-three rows and named, for each Millennium row, its target bit, its even register and its gate typing. Nothing Escapes Least Erasure (Islam 2026j) graded every row on two channels: the kinetic channel, where the value holds on the class of instances that exist, and the formal channel, the traditional route, blocked wherever every derivation from the register's resources is refused, which its Odd-Supply Block closes by theorem on every open row that carries a registration and an offering from outside the register can cross. This paper reports the seven closure papers whose first editions the author had published by 5 October 2026, each in its edition of record, dated 6 October 2026, and reads them as the arc's results, row by row, on both channels. Two senses of *formal* are kept apart throughout: the title's *Formal Closure* is each row's closure formalized in Lean, the kernel theorem from the act, the closing theorem compiled with its axiom cone printed; the *formal channel* is the traditional route of derivation from the register, blocked by theorem on six of the seven Millennium rows beyond their certified parts and crossed on Poincaré alone, by Perelman's offering, at the grade of its citation (Perelman 2002), full grade in the carried ledger of Islam (2026j). The precedence rule of this ledger is fixed once: a row's closure paper governs its row over the carried ledger of Islam (2026j), whatever the edition dates, and where the two differ the departure is a rule of this ledger, not a theorem that supercedes that volume's.

2. METHOD

Each row is recorded on the columns of one cumulative ledger, the twenty-three rows of Islam (2026j, Book III) with the closure papers placed on top, each governing its row over that carried ledger whatever

the edition dates. The **target bit** and **even register** are those of Islam (2026b, Table 2), cited there and not reprinted in Table 1. The **block bit** is the paper's own block theorem, the formalized statement that nothing given decides the row. The **hypothesis bit** is the located seat and its single missing orientation where the kernel states a seat; on Navier–Stokes and Yang–Mills, where no seat is formalized, the one bit, on Yang–Mills one per part, carried through the record wall; and on P versus NP the seat proved vacant and the keyed separation bit, which no reading of the vanished seat's record returns. The **resolution bit** is what the paper supplies by the act, existence read on the row, at the seat where its kernel states one and on the row where it states none. The **channels** are those of Islam (2026j): the kinetic channel is crossed where the value holds on the actuated class, and the formal channel is blocked by theorem wherever every derivation from the register's resources is refused (the Odd-Supply Block; `barrier_unconditional`, in the row kernel of Islam 2026j, on every row with a registration; on row 1, whose locus is vacant, the P versus NP kernels' own blocks: `record_decides_nothing` and `keyless_forces_nothing` in `PNP_Least_Escape.lean`, `block_chain` and `locus_vanishes` in `PNP_Frontier_II.lean`). **What remains** is read from the closure paper of each row. Each kernel is cited by its receipts as printed in its paper. The row kernels were recompiled in this programme on Lean 4.19.0 and 4.22.0: the Riemann kernel, 394 theorems, its closing theorem `rh_from_the_act` on no axiom and 183 of the 394 on no axiom by the compiled census of its Appendix B.3, 102 of them pinned under `#guard_msgs`; the four P versus NP kernels, `PNP_Root.lean`, `PNP_Least_Escape.lean`, `PNP_Fortification.lean` and `PNP_Frontier_II.lean`, 7, 46, 21 and 142 theorems, the closing theorem `sep_from_the_act`, in `PNP_Least_Escape.lean`, on no axiom; and the Navier–Stokes, Hodge, Yang–Mills, Goldbach, Poincaré and Birch and Swinnerton-Dyer kernels, 65, 77, 70, 104, 80 and 101 theorems, every theorem on no axiom, the arithmetic bridge of the Goldbach kernel included (Islam 2026l), its remainder read from the definition of % itself. Each row's lock kernel binds the root by its act, the root read on its row, its keyed act, its closure and its record wall in one theorem, the `_lock`: the Riemann kernel, the one kernel of each of the six rows named last, and for P versus NP `PNP_Least_Escape.lean`, the other three P versus NP kernels being its fortification, frontier and root (`PNP_Fortification.lean`, `PNP_Frontier_II.lean`, `PNP_Root.lean`); and every paper carries the root itself in its Appendix R.

3. RESULTS

3.1 The seven completed papers

The Navier–Stokes closure (Islam 2026c) proves that existence as given holds in the smooth and the blowup

world alike, and that existence read on the row, the act, is regularity, from which global regularity follows by one act; on matter it proves that no flow blows up from the quantum speed limit in two senses, in the limit sense from the speed limit alone, on a world carrying the speed limit and nothing else (`no_limit_blowup_from_speed_only`) and on the physical frame (`no_limit_blowup_from_speed`), and in the count sense from the speed limit with the cascade, a hypothesis carried as a field, the limit-form link between the count and the stages, not a cited physical result and not a definition of blowup (`no_actual_blowup`), both fields load-bearing for the count sense (`speed_is_load_bearing`, `cascade_is_load_bearing`). The Hodge closure (Islam 2026d) proves the act equal to algebraicity of every Hodge class and seals the parts proved in dimension at most three and on the outer codimensions in every dimension (`low_dim_decided`, `outer_decided`), the middle range $2 \leq p \leq n - 2$ standing on the act from dimension four (`middle_not_forced`, `every_middle_codimension_unforced`). The Yang–Mills closure (Islam 2026e) divides the row into its two halves, the theories built and the gap, and proves that the built half does not give the gap (`existence_does_not_give_gap`: `ExistsAll` without `GapAll`), while the act, constructed and gapped, is the value (`act_is_the_value`); it seals the lattice value of the strong-coupling sub-frame on its own substrate (`strong_coupling_value`), which no theorem carries to the continuum frame, where the value stands on the act, and reads its matter face on the pure-gauge sector, on the confinement record, the area law (Creutz 1980), and the premise that a massless confined excitation carries colour, at premise grade, its refuter a massless colour singlet (`premise_excludes_singlet`). The Poincaré closure (Islam 2026f) scopes the row to three-manifolds and proves that Perelman’s witness, in the shape the kernel types, strict descent off the seat, supplies the act and that the act is wider than the witness, holding where no witness exists (`Witness`, `witness_gives_act`, `act_without_witness`, `crossing_types_the_crossing`). The Riemann closure (Islam 2026g) closes the hypothesis to one named bit by one act of least erasure. The closure of computational separation (Islam 2026h) closes P versus NP on existence read on computation. The Birch and Swinnerton-Dyer closure (Islam 2026i), whose kernel carries the functional equation as the reflection law of its centred expansions, the field reflect of `Centred`, cited, proves that a minus sign forces vanishing at the centre and odd order on every signed frame (`sign_fixes_curve_parity`, on `SignedFrame`), and a plus sign even order (`plus_order_even`, on `Centred`), that every first-order reading is silent above order one, and that the act is the weakest premise of the rank part (`act_is_the_weakest_premise`) and the only one its closing theorem consumes (`bsd_from_existence`, on `Actual-`

`Curves`), the full act, strictly stronger, the only premise the full closure consumes (`full_act_is_the_weakest_premise`, `bsd_full_from_existence`, on `ActualCurves-Full`).

3.2 The cumulative ledger

The block bit is complete on all seven rows. Every paper proves, in its kernel, that no statement holding on every frame equals the row’s value, and six name the traditional barrier their block formalizes; the Riemann row’s, the Davenport–Heilbronn register, is named in Islam (2026b, Table 2).

The hypothesis bit is settled on all seven rows. The Riemann, Poincaré and Birch and Swinnerton-Dyer papers locate the seat and its single missing orientation in their kernels (`fold_fixed_iff`, `seat_is_fixed`, `Centred`), as the Goldbach paper does for its row (`mirror_has_one_seat`) and the Hodge paper now does (`hodge_seat`: conjugation of Hodge types, fixed on the diagonal); the Navier–Stokes and Yang–Mills papers formalize no seat and carry the one bit through the record wall, on Yang–Mills one per part, existence and the gap, both values of each bit sharing one record (`record_wall`, `fibre_is_two`), the Yang–Mills seat typed none formal in Islam (2026j); the P versus NP paper proves its complementation seat vacant (`seat_vacant`, `al_no_seat`, in `PNP_Frontier_II.lean`), so that on that row no orientation sits at a seat, and the hypothesis bit is the keyed separation bit, which no reading of the vanished seat’s record returns (`separation_keyed`, `vanished_seat_reads_no_machine`, in the same kernel). Where the row carries a parity, the series proves it: the BSD kernel proves that the minus sign forces odd order on every signed frame (`sign_fixes_curve_parity`, on `SignedFrame`) and a plus sign even order (`plus_order_even`, on `Centred`), and that rank–order parity agreement (`ParityValue`) is one bit and not the value (`parity_is_not_the_value`).

The resolution bit is supplied on all seven rows by the act, and nothing remains on any row’s closure, the Navier–Stokes row read as its unforced Millennium line 3a, the forced line 3b being its own entry. On Poincaré and on BSD at orders zero and one, the corpus also holds a dated construction, Perelman’s in 2002 and Gross–Zagier with Kolyvagin in 1986 and 1990; the series types Perelman’s as a witness that, in the shape the kernel types, supplies the act, the act wider than the witness and holding where no witness exists (`Witness`, `witness_gives_act`, `act_without_witness`, `crossing_types_the_crossing`), and Gross–Zagier with Kolyvagin as a rank-only witness of the act at orders at most one, which does not give the leading coefficient (`low_witness_not_full`); the full row, rank and leading coefficient, rests on the full act (`bsd_full_from_existence`), strictly stronger than the act (`full_act_strictly_stronger`). On row 1 this ledger departs from the carried ledger of Islam (2026j) by the act

#	Row	Paper (Zenodo)	Block bit (traditional barrier)	Hypothesis bit (seat · one orientation)	Resolution bit (closure paper)	Kinetic channel	Formal channel (traditional route)	What remains	Physical witness · grade · refuter
1	P vs NP	23133632	relativization; natural proofs; algebrization	seat vacant: complementation fixes no bit; the keyed separation bit, returned by no reading of the record	the act: existence read on computation; crossed by the act where Islam (2026j) and the kernel's own seat token print the dot and its verdict with the act supplied is the separation seal, a rule of this ledger	crossed, premise grade	blocked by theorem (the kernel's own blocks)	nothing on the closure	no physical polynomial NP solver registered (Aaronson 2005); Landauer floor · corroboration · refuter: one such solver
2	Riemann	23034066	Davenport-Heilbronn; zero-free regions short of the line	the fold, fixed set the critical line; one bit, least erasure	the act: least erasure at the actual zero set	crossed, premise grade; 3×10^{12} certified	blocked by theorem	nothing on the closure	energy carrier $s = \frac{1}{2} + iE$; GUE pair correlation; Lee-Yang circle theorem (Islam 2026j) · structural · no physical refuter; one computed zero off the line, a mathematical refuter (F-Computed)
3a	Navier-Stokes, unforced	22670344	supercriticality; Tao's averaged blowup	no seat formalized; one bit through the record wall, fibre two	the act at the continuum; speed-limit theorem on matter	crossed on actual flows; matter, a theorem from the cited speed limit, alone in the limit sense and with the cascade, a hypothesis carried as a field, in the count sense	blocked by theorem	nothing on the closure	every physical frame regular (matter_is_regular), an actual fluid read as a physical frame by the reader's identification · theorem from the cited speed limit with the cascade, a hypothesis carried as a field · refuter: one finite-time singularity of an actual unforced flow
3b	Navier-Stokes, forced	—	forced breakdown not transported to closed systems	the scaling group, fixed locus the critical quantities (Islam 2026j)	—	—	blocked by theorem	forced construction under audit; one order-type	none
4	Yang-Mills	23162231	no 4D continuum construction; gap only at strong coupling	no seat formalized (Islam 2026j: none formal); one bit per part, existence and the gap, through the record wall, fibre two	the act: constructed and gapped	crossed, premise grade; the matter face on the pure-gauge sector, premise grade	blocked by theorem	nothing on the closure	confinement record of pure lattice Yang-Mills, the area law (Creutz 1980), no free colour; glueballs near 1.7 GeV (Morningstar and Peardon 1999) · record with the colour premise, premise grade · refuter: one massless colour singlet, a theory without a mass floor none: the paper carries no physical witness; flux quantization corroborates the integral shape, not the row's value
5	Hodge	22701510	integral classes fail (Atiyah-Hirzebruch)	no seat formalized; one bit through the record wall, fibre two	the act: every class realized; codim 1, dim ≤ 3 sealed	crossed, premise grade	crossed in dim ≤ 3 and on the outer codimensions in every dim; blocked in the middle range $2 \leq p \leq n - 2$ from dim 4	nothing on the closure	none · computational: 389a1 regulator 0.1524601779
6	BSD	23164992	Heegner class torsion above order one	the centre of the functional equation, fixed by the reflection; one bit, the rank on the seat	the full act, read on the full row: rank and leading coefficient, strictly stronger than the act; Gross-Zagier-Kolyvagin a rank-only witness at orders ≤ 1	crossed, premise grade	rank part crossed at orders ≤ 1 ; leading coefficient not crossed by it; blocked above	nothing on the closure	none · computational: 389a1 regulator 0.1524601779
7	Poincaré	23162240	rescaling reading class before 2002	the round metric, fixed under the flow; one bit, rest on the seat	the act: Perelman's witness, in the shape the kernel types, a supply of it	crossed; discharged, at the grade of its citation (Perelman 2002)	crossed, at the grade of its citation (Perelman 2002), full grade in the carried ledger of Islam (2026j)	nothing	Ricci flow as RG flow (Friedan); W as entropy · structural
8	Goldbach	23162238	no proof beyond the certified region	the mirror $p \leftrightarrow n - p$, one seat; one bit, the split	the act at the mirror's seat	crossed; certified to 4×10^{18}	blocked by theorem	nothing on the closure	certificate computations, each erased bit costing at least $k_B T \ln 2$ · kinetic registration
9	Twin primes	—	sieve parity problem	the pair swap; one bit	—	one witness registered	blocked by theorem	infinitude with sieve mass	none
10	Legendre	—	gap bounds short of $\sqrt{n}$	the mirror about the interval's centre; one bit	—	registered to 2×10^9	blocked by theorem	a gap bound below $2\sqrt{n}$	certificate computation
11	abc	—	no certified region	no seat	—	no cut	blocked by theorem	the inequality whole	none
12	Jacobian	—	degree ≤ 2 only; no global seat	local seat only	—	degree ≤ 2 proved	blocked by theorem	invertibility of cubic maps	none
13	Mersenne	—	Π_2	no seat	—	one witness registered	blocked by theorem	infinitude	none
14	Odd perfect	—	Π_1	fixed points of the aliquot map, not an involution	—	none below 10^{1500}	blocked by theorem	nonexistence above the height	certificate computation
15	Smooth 4D Poincaré	—	Whitney trick fails	no seat	—	topological case proved	blocked by theorem	a smooth classification	none
16	Collatz	—	Π_2	no seat	—	proved to 2^{68} registered	blocked by theorem	termination for all n	certificate computation
17	Hadwiger	—	no global seat	local seat only	—	$t \leq 6$ proved	blocked by theorem	the bound at $t \geq 7$	none
18	Sunflower	—	Σ_2 ; no cut	no seat	—	no cut	blocked by theorem	the constant-base bound	none
19	Erdős-Straus	—	no global seat	local seat only	—	to 10^{17} registered	blocked by theorem	the representation for all n	certificate computation
20	Beal	—	no global seat	local seat only	—	searched boxes	blocked by theorem	nonexistence outside every box	none
21	Invariant subspace	—	general bounded operator	the adjoint, fixed locus the self-adjoint operators; one bit	—	normal, compact, finite-dim proved	blocked by theorem	every quantum observable on the proved locus; tail kinetically empty	every quantum observable on the proved locus; tail kinetically empty
22	Hilbert's 16th	—	no uniform bound	no seat	—	per-field finiteness proved	blocked by theorem	H(2)	oscillators instantiate finiteness, never a uniform bound
23a	Pair correlation	23080221	restricted range only, under RH	RH's fold; one bit	the act (NELE)	crossed on chaotic spectra and computed zeros (Odlyzko 1987)	blocked by theorem	nothing on the closure	time-broken chaotic spectra in the unitary class · measured · refuter: one off the class
23b	Lindelöf	—	proved exponents only	RH's fold; the exponent ϵ , one bit each	—	proved exponents	blocked by theorem	the exponent ϵ	none

Table 1 | The cumulative ledger. The twenty-three rows and their numbering are those of Islam (2026j, Book III); the seven closure papers and the Goldbach paper are placed on top of that ledger, and a row's closure paper governs its row over the carried ledger whatever the edition dates, this ledger departing from Islam (2026j) on rows 1, 5, 6 and 8 by the act. The block bit is complete on all seven Millennium rows. The hypothesis column prints the seat and its one orientation: from the row kernel on rows 2, 5, 6, 7 and 8; on rows 3a and 4 no seat is formalized and the one bit, on row 4 one per part, is carried through the record wall; on row 1 the closure paper proves the complementation seat vacant, and the hypothesis bit is the keyed separation bit, which no reading of the vanished seat's record returns; on the remaining rows the cell is the carried ledger's, a global seat where it exhibits one, and on rows 12, 17, 19 and 20 a local seat only. Where the act is supplied, the kinetic channel holds at the grade of the act, on Poincaré discharged by Perelman's witness at the grade of its citation, a theorem of the literature carried as the witness in the shape the kernel types, the kernel constructing no witness; elsewhere its cell records the classes on which the value is proved; the formal channel is the traditional route, blocked by theorem on every open row. On rows with no closure paper, "what remains" is the tail as Islam (2026j, Table 2) states it. Every verdict of the seat survives every rescaling, so no magnitude is read; measured magnitudes corroborate their cited formulas (Book VII, with its sixty-four typed falsifiers).

alone: the row is crossed by the act, the closure of its own paper (`sep_from_the_act`, in `PNP_Least_Escape.lean`, Islam 2026h), where the computed status of Islam (2026j) is the dot (`dot_is_final`, in the row kernel of Islam 2026j); the P versus NP kernels' own tokens carry both readings: the seat token of the vanished seat is the dot (`tongue_face`, in `PNP_Frontier_II.lean`: `seatToken groundDim = .dot`), and the row's verdict with the act supplied is the separation seal and without it the dot (`value_marked_dot`, in `PNP_Least_Escape.lean`: `verdict true = .sealSep, verdict false = .dot`); this ledger records the act supplied, and the departure is a rule of this ledger, not a theorem that supersedes theirs.

3.3 The physical-witness column

The last column of Table 1 records what nature has constructed at each row, its grade, and the one observation that would refute it. Nature supplies a provenance-clean witness on Navier–Stokes matter, a theorem from the cited speed limit, alone in the limit sense and with the cascade, a hypothesis carried as a field, in the count sense; on Yang–Mills a record with a premise, the confinement record of the pure-gauge sector and the premise that a massless confined excitation carries colour, at premise grade; measured on pair correlation; structural on Riemann and Poincaré, carried from Islam (2026j); by registration on the height-cut rows, whose certified regions were computed, each erased bit costing at least $k_B T \ln 2$; and nothing on Hodge, whose paper carries no physical witness, flux quantization corroborating the integral shape and not the row's value.

4. DISCUSSION

4.1 Two channels, one ledger

The results agree on the first two bits without remainder, and the third bit is graded on the two channels of Islam (2026j). On the kinetic channel the act crosses each row on the class of instances that exist, at the grade of the act, premise grade, on the root beneath it: the root, proved at theorem grade with no hypothesis on its constructed domain and as the law of every root that grounds itself (Appendix R), universally presupposed and undeniable in act, every denial of it an act that instances it (`denial_reenacts_root`, in Appendix R and in every row kernel). On the rows Islam (2026j) crosses by a carrier, the value on the actuated class is discharged by that carrier and coverage is the one premise, posited once at premise grade (`rest_family`, in the row kernel of Islam 2026j; the carried kernel's comment names that premise the root; it is the coverage premise, not the root of Appendix R); Poincaré, crossed there by a formal offering with no world carrier, carries in this ledger the act at premise grade, discharged by Perelman's witness at the grade of its citation, a theorem of the literature carried as the witness, in the shape the Poincaré kernel types (`witness`, con-

sumed by `witness_gives_act`), the kernel constructing no witness, so that the row stands at the grade of that citation (crossed; discharged, at the grade of its citation); on rows 1, 5, 6 and 8, which that volume leaves the dot or pending, the premise is the act field of the row kernel; on every row this ledger records that field, one keyed field per row. Holding in every world, carried by every act, the root decides no value by itself; read on each row, with the row's existents and its actuation, it is the act, keyed, which decides it (`root_read_on_row_is_keyed`, in each row kernel). On the formal channel the traditional route, every derivation from the register's resources, is blocked by theorem on every open row (the Odd-Supply Block; `barrier_unconditional`, Islam 2026j, on every row with a registration; on row 1, whose locus is vacant, the P versus NP kernels' own blocks: `record_decides_nothing` and `keyless_forces_nothing` in `PNP_Least_Escape.lean`, `block_chain` and `locus_vanishes` in `PNP_Frontier_II.lean`); the volume states the grade itself, that the act is the hypothesis supplied at the actual instance set and not a result unconditional in the sense of the traditional route. Poincaré is the one row crossed on the formal channel, by Perelman's offering, at the grade of its citation (Perelman 2002), the Poincaré kernel's formal route consuming the cited dimension-three field of `Dimensional` (`every_dimension_decided, three_is_load_bearing`), full grade in the carried ledger of Islam (2026j), whose row kernel records the shape of that event: an offering with mass from outside the register crosses on arrival (`one_offering_crosses`).

4.2 What each channel claims

The two channels never contradict, because they make different claims. The kinetic channel proves the value on every instance that exists, at the act's grade, and the closure paper of each row leaves nothing remaining on that closure. The formal channel records exactly what the traditional route can reach, which on every open row is nothing beyond the certified parts, by theorem. A reader who asks whether a row is derived on the traditional route receives the formal channel's answer, blocked by theorem; a reader who asks whether the row is closed receives the kinetic channel's, closed by one act at its grade.

4.3 The two registers

Status

Unconditional, at theorem grade. The root every row shares, on its constructed domain and for every root that grounds itself, on no axiom (Appendix R). On each row's kernel: the block bit; the seat and its one orientation on the Riemann, Hodge, Poincaré and Birch and Swinnerton-Dyer rows, the vacant seat on P versus NP, and on Navier–Stokes and Yang–Mills the one bit, on Yang–Mills one per part, through the record wall with no seat formalized; the identity of the root read on the row with the value (`root_read_`

on_row_is_keyed, in each row kernel; root_on_row_is_the_value, in the Navier–Stokes, Hodge, Yang–Mills, Goldbach, Poincaré and Birch and Swinnerton-Dyer kernels); the closure from the reading; and the lock, at the footprint each kernel prints, on no axiom or on Lean’s standard axioms alone.

On the reading, at premise grade. The reading itself, the root read on the actual frame of each row, and with it the value of each of the seven rows, supplied by the act, existence read on the row; on Poincaré that premise is discharged by Perelman’s witness at the grade of its citation, a theorem of the literature carried as the witness, in the shape the kernel types (Witness, consumed by witness_gives_act), the kernel constructing no witness, and the row stands at the grade of that citation.

Everything except the resolution bit and the fields each row’s paper names, cited or carried as hypotheses (the colour premise of the Yang–Mills matter face, the cascade of the Navier–Stokes matter theorem), is proved with no posit: the floor on no axiom at all, each row’s closure on the footprint its kernel prints, on no axiom or on Lean’s standard axioms alone; and the resolution bit of each row, its reading supplied on the actual instances, is all that remains on each row’s closure, at premise grade, on Poincaré discharged by Perelman’s witness at the grade of its citation, the kernel constructing no witness.

4.4 What the series has established

On either reading the series has done what no prior body of work did on these rows together: it formalized each row’s block in checkable kernels, located the seat and its single orientation on every row whose kernel states one and carried the one bit, on Yang–Mills one per part, through the record wall on the two that formalize none, and typed every historical crossing against one schema. The barrier literature of each row and the dated crossings of two of them now sit in one vocabulary, and each row’s remaining entry is named.

5. FALSIFIERS

F-Block. A statement true on every frame of a row’s kernel that equals the row’s value; it refutes that row’s block bit.

F-Seat. A row whose corpus-compatible target set is shown to exceed two at its seat; it refutes the one-bit typing of that row (Islam 2026b, Section 4.1). On row 1 F-Seat is vacuous: the seat is vacant (seat_vacant, in PNP_Frontier_II.lean), and no target set stands at it; on rows 3a and 4, where no seat is formalized, F-Seat has no target set either, the one bit there, on row 4 one per part, carried through the record wall (record_wall, fibre_is_two).

F-Entry. A provenance-clean construction entering at an open seat and deciding the row against the act; it refutes that row’s closure.

F-World. One physical refuter per kinetically crossed row that Table 1 prints with one: one poly-

nomial NP solver registered in nature, one finite-time singularity of an actual unforced flow, one massless colour singlet, one time-broken chaotic spectrum off the unitary class (Islam 2026j). The Riemann, Hodge, Birch and Swinnerton-Dyer, Poincaré and Goldbach rows have no physical refuter: the Riemann row’s refuter is mathematical, one computed zero off the line (F-Computed, Islam 2026g); Hodge and Birch and Swinnerton-Dyer carry no physical witness; Poincaré carries a structural one and Goldbach a registration of computed certificates. On these five rows F-Block and F-Entry are the refuters, and on the Goldbach row one computed even number with no split refutes the act, as its paper proves (Islam 2026l).

6. CONCLUSION

Block, hypothesis and resolution: one bit, one bit and one bit, irreducible and in that order, on all seven Millennium rows, the hypothesis bit on P versus NP the keyed separation bit at no seat. The seven completed papers carry the first two bits on all seven Millennium rows and supply the third as the act, closing each row on the kinetic channel with nothing remaining on its closure. The formal channel, the traditional route, is blocked by theorem on every open row beyond its certified parts, and Poincaré alone is crossed on it, by Perelman’s offering, at the grade of its citation (Perelman 2002), full grade in the carried ledger of Islam (2026j).

APPENDIX R • THE ROOT ON NO AXIOM AND NO POSIT

The root on which this paper stands is proved here, in this paper, so that no reader needs another document to check it. The kernel below, R.1, T0E_Zero.lean, is carried verbatim from the root paper (Islam 2026k). Compiled on Lean 4.19.0 and on Lean 4.22.0 it exits 0 with no message: it declares no axiom, imports nothing, and every one of its fifteen theorems prints *does not depend on any axioms*, pinned at the foot of the listing. Five declarations carry its root sections, SelfGrounding, SelfVerifying, ΔE_0 , RA_0 and ra_0 , and all five stand in the listing; the algebra of its master seal rests on the remaining definitions of the listing, the quaternion chart among them. A root grounds itself when deeds occur and every deed instances it (SelfGrounding; the root kernel’s own comments call a deed an act). The constructed domain has one existent, whose energy of actuation is one, so the Root Axiom there reads $\forall x, 0 < \Delta E_0(x) (RA_0)$. On that domain the Root Axiom is a theorem with no axiom and no hypothesis (root_on_the_constructed_domain). For every root that grounds itself, the root holds, every deed re-enacts it, and no outside proof adds anything to it, one closed theorem on no axiom (the_floor_is_universal). Every denial of the root is an act that instances it (denial_reenacts_root); the root is held by its deed it-

self (seated_undeniable); it is self-verifying (denial_instantiates); and no level stands above it (no_level_above). The master seal binds the root's universal law and the constructed root with the Return, the scalar line as the fixed set of conjugation on the chart and the twelve gates of the quaternions in one theorem on no axiom (the_master_seal). The body of this paper reads the root on its row; this appendix is the root itself. SHA-256: 0cfcd9f6b399ecbaef01e762a4d9f10490288f1053827b404428062e55076238.

```

/-
TOE_Zero.lean · Theory of Theories of Everything (TOE of All
TOEs): Existence Proves Existence Only by Motion
The zero-axiom, zero-posit kernel. Core Lean 4: no import, no
axiom declared, no sorry, no native_decide, and
every theorem rests on no axiom at all, not even propositional
extensionality, quotients or choice.
I · The self-grounding root. Acts occur and every act instances
the root. Then every denial is an act and
re-enacts the root, the root is held by the act itself, and no
external proof adds anything to it: the
Omega Boundary, proved for every root that grounds itself,
universally, by quantification.
II · The constructed root. On the constructed domain the Root
Axiom is a theorem.
III · The Return and the line. On the integer quaternions i·j·k =
-1 and each unit squares to -1; conjugation is
an involution and its fixed set on the chart is exactly the
scalar line.
IV · The twelve gates, recovered three ways inside the integers of
the quaternions: the twenty-four Hurwitz units
counted twelve up to sign; the twelve even permutations with
class equation 1, 3, 4, 4; and the norm-two
shell of twenty-four, split twelve and twelve.
V · The regress terminated. Above a self-grounding root no level
grounds it further.
VI · The master seal, every result bound in one theorem.
-/
namespace TOE0

/-! ## I · The self-grounding root: the Omega Boundary -/

/-- A self-grounding root: acts occur, and every act instances the
root. -/
structure SelfGrounding (R : Prop) where
  Act      : Type
  anAct     : Act
  instances : Act → R

/-- A proposition whose denial hands it over. -/
def SelfVerifying (P : Prop) : Prop := ¬P → P

/-- THE OMEGA BOUNDARY: every denial of the root is an act, and
re-enacts it. -/
theorem denial_reenacts_root {R : Prop} (G : SelfGrounding R)
  (denial : G.Act) : R := G.instances denial

/-- EXISTENCE PROVES EXISTENCE ONLY BY MOTION: the root is held by
the act itself. -/
theorem seated_undeniable {R : Prop} (G : SelfGrounding R) : R :=
  G.instances G.anAct

/-- No external proof adds anything to the root. -/
theorem external_proof_adds_nothing {R : Prop} (G : SelfGrounding R)
  (Q : Prop) : (Q → R) ↔ R :=
  (fun _ => G.instances G.anAct, fun r _ => r)

/-- The deed of denying the root hands it over. -/
theorem denial_instantiates {R : Prop} (G : SelfGrounding R) :
  SelfVerifying R := fun _ => G.instances G.anAct

/-- THE FLOOR IS UNIVERSAL: for every root that grounds itself, the
root holds, every act re-enacts it, and no
external proof adds to it. -/
theorem the_floor_is_universal :
  ∀ (R : Prop) (G : SelfGrounding R), R ∧ (∀ _ : G.Act, R) ∧ (∀ Q :
  Prop, (Q → R) ↔ R) :=
  fun _ G => (seated_undeniable G, denial_reenacts_root G,
  external_proof_adds_nothing G)

/-! ## II · The constructed root -/

/-- The Root Axiom on the constructed one-point domain: its one
existent actuates, at positive energy. -/
def ΔE₀ : Unit → Int := fun _ => 1
def RA₀ : Prop := ∀ x : Unit, 0 < ΔE₀ x

/-- The constructed domain grounds its root: its one act is a deed,
and the deed actuates. -/
def ra₀ : SelfGrounding RA₀ := (Unit, (), fun _ _ => show (0 : Int)
  < 1 by decide)

/-- THE ROOT ON THE CONSTRUCTED DOMAIN, A THEOREM. -/
theorem root_on_the_constructed_domain : RA₀ := seated_undeniable
  ra₀

/-! ## III · The Return and the line -/

structure Q where
  (a b c d : Int)
  deriving DecidableEq

def Q.mul (x y : Q) : Q :=
  (x.a * y.a - x.b * y.b - x.c * y.c - x.d * y.d, x.a * y.b + x.b *
  y.a + x.c * y.d - x.d * y.c,
  x.a * y.c - x.b * y.d + x.c * y.a + x.d * y.b, x.a * y.d + x.b *
  y.c - x.c * y.b + x.d * y.a)

def Q.conj (x : Q) : Q := (x.a, -x.b, -x.c, -x.d)
def qone : Q := (1, 0, 0, 0)
def qneg : Q := (-1, 0, 0, 0)
def qi : Q := (0, 1, 0, 0)
def qj : Q := (0, 0, 1, 0)
def qk : Q := (0, 0, 0, 1)

/-- THE RETURN: i·j·k = -1. -/
theorem the_return : (qi.mul qj).mul qk = qneg := by decide

/-- Each unit squares to -1. -/
theorem units_square_to_minus_one : qi.mul qi = qneg ∧ qj.mul qj =
  qneg ∧ qk.mul qk = qneg := by decide

/-- The chart: every quaternion with coordinates in -2 ... 2. -/
def chart : List Int := [-2, -1, 0, 1, 2]
def quats : List Q :=
  chart.flatMap fun a => chart.flatMap fun b => chart.flatMap fun c
    => chart.map fun d => (a, b, c, d)

set_option maxRecDepth 100000 in
/-- Conjugation is an involution on the chart. -/
theorem conj_involution_on_the_chart : quats.all (fun q =>
  q.conj.conj == q) = true := by decide

set_option maxRecDepth 100000 in
/-- THE LINE: conjugation fixes exactly the scalar line on the
chart. -/
theorem the_line_is_the_fixed_set :
  quats.all (fun q => (q.conj == q) == (q.b == 0 && q.c == 0 &&
  q.d == 0)) = true := by decide

/-! ## IV · The twelve gates, recovered three ways -/

/-- Hurwitz quaternions in doubled coordinates: all even, or all
odd. -/
def hurwitz (a b c d : Int) : Bool :=
  (a % 2 == 0 && b % 2 == 0 && c % 2 == 0 && d % 2 == 0) || (a % 2 !=
  0 && b % 2 != 0 && c % 2 != 0 && d % 2 != 0)

/-- Doubled coordinates -2 ... 2 hold every unit (doubled norm four)
and every element of the norm-two shell (doubled
norm eight), since a doubled coordinate of absolute value three
or more already exceeds both norms. -/
def doubled : List (Int × Int × Int × Int) :=
  chart.flatMap fun a => chart.flatMap fun b => chart.flatMap fun c
    => chart.map fun d => (a, b, c, d)

/-- The Hurwitz units: norm one, doubled norm four. -/
def units : List (Int × Int × Int × Int) :=
  doubled.filter fun (a, b, c, d) => a * a + b * b + c * c + d * d ==
  4 && hurwitz a b c d

/-- The first nonzero coordinate is positive: one representative of
each pair ±u. -/
def upToSign (p : Int × Int × Int × Int) : Bool :=

```

```

let (a, b, c, d) := p
if a != 0 then 0 < a else if b != 0 then 0 < b else if c != 0 then
  0 < c else 0 < d

set_option maxRecDepth 100000 in
/-- FIRST WAY: twenty-four Hurwitz units, twelve up to sign. -/
theorem twenty_four_units_twelve_gates : units.length = 24 ∧
  (units.filter upToSign).length = 12 := by decide

/-- The norm-two shell: doubled norm eight. -/
def shell : List (Int × Int × Int × Int) :=
  doubled.filter fun (a, b, c, d) => a * a + b * b + c * c + d * d ==
    8 && hurwitz a b c d

set_option maxRecDepth 100000 in
/-- THIRD WAY: the norm-two shell has twenty-four elements, split
  twelve and twelve by sign. -/
theorem the_shell_splits_twelve_and_twelve :
  shell.length = 24 ∧ (shell.filter upToSign).length = 12 ∧
  (shell.filter (fun p => !upToSign p)).length = 12 := by
  decide

/-- Permutations of four points, as their lists of images. -/
def perms4 : List (List Nat) :=
  [0, 1, 2, 3].flatMap fun a => [0, 1, 2, 3].flatMap fun b => [0, 1,
    2, 3].flatMap fun c =>
    [0, 1, 2, 3].filterMap fun d => if [a, b, c, d].eraseDups.length
      == 4 then some [a, b, c, d] else none

def inversions (p : List Nat) : Nat :=
  ((List.range 4).flatMap fun i => (List.range 4).map fun j => (i,
    j)).countP
  fun (i, j) => i < j && p.getD j 0 < p.getD i 0

/-- The twelve gates as the even permutations. -/
def gates : List (List Nat) := perms4.filter fun p => inversions p %
  2 == 0

def compose (p q : List Nat) : List Nat := q.map fun i => p.getD i 0
def inverse (p : List Nat) : List Nat := [0, 1, 2, 3].map fun i =>
  p.idxOf i

def conjClass (g : List Nat) : List (List Nat) := (gates.map fun h
  => compose (compose h g) (inverse h)).eraseDups

def insertSorted (n : Nat) : List Nat → List Nat
| [] => [n]
| m :: ms => if n ≤ m then n :: m :: ms else m :: insertSorted n ms

def sortNat (l : List Nat) : List Nat := l.foldr insertSorted []

/-- The class equation from the sorted class size of each element: a
  class of size s contributes s elements. -/
def collapse : Nat → List Nat → List Nat
| 0, _ => []
| _ + 1, [] => []
| fuel + 1, s :: rest => s :: collapse fuel (rest.drop (s - 1))

set_option maxRecDepth 100000 in
/-- SECOND WAY: the twelve gates, with class equation 1, 3, 4, 4. -/
theorem the_class_equation :
  gates.length = 12 ∧ collapse 12 (sortNat (gates.map fun g =>
    (conjClass g).length)) = [1, 3, 4, 4] := by
  decide

/-! ## V · The regress terminated -/

/-- THE REGRESS ENDS AT THE ROOT: any proposition offered as a
  further ground of the root is idle; the root needs
  no level above it. -/
theorem no_level_above {R : Prop} (G : SelfGrounding R) : ∀ Q :
  Prop, (Q → R) → R := fun _ => G.instances G.anAct

/-! ## VI · The master seal -/

/-- THE MASTER SEAL: the floor is universal; the root is a theorem
  on the constructed domain; the Return closes;
  the line is the fixed set; and the twelve gates are recovered
  three ways. -/
theorem the_master_seal :
  (∀ (R : Prop) (G : SelfGrounding R), R ∧ (∀ _ : G.Act, R) ∧ (∀ Q
    : Prop, (Q → R) ↔ R)) ∧
  RA₀ ∧ (qi.mul qj).mul qk = qneg A

```

```

quats.all (fun q => (q.conj == q) == (q.b == 0 && q.c == 0 &&
  q.d == 0)) = true ∧
  (units.filter upToSign).length = 12 ∧ gates.length = 12 ∧
  (shell.filter upToSign).length = 12 ∧
  collapse 12 (sortNat (gates.map fun g => (conjClass g).length))
    = [1, 3, 4, 4] :=
  (the_floor_is_universal, root_on_the_constructed_domain,
    the_return, the_line_is_the_fixed_set,
    twenty_four_units_twelve_gates.2, the_class_equation.1,
    the_shell_splits_twelve_and_twelve.2.1, the_class_equation.2)

end TOE0

/-! ## The zero law, pinned: every cone is empty -/
/-- info: 'TOE0.denial_reenacts_root' does not depend on any axioms
  -/
#guard_msgs in #print axioms TOE0.denial_reenacts_root
/-- info: 'TOE0.seated_undeniable' does not depend on any axioms -/
#guard_msgs in #print axioms TOE0.seated_undeniable
/-- info: 'TOE0.external_proof_adds_nothing' does not depend on any
  axioms -/
#guard_msgs in #print axioms TOE0.external_proof_adds_nothing
/-- info: 'TOE0.denial_instantiates' does not depend on any axioms
  -/
#guard_msgs in #print axioms TOE0.denial_instantiates
/-- info: 'TOE0.the_floor_is_universal' does not depend on any
  axioms -/
#guard_msgs in #print axioms TOE0.the_floor_is_universal
/-- info: 'TOE0.root_on_the_constructed_domain' does not depend on
  any axioms -/
#guard_msgs in #print axioms TOE0.root_on_the_constructed_domain
/-- info: 'TOE0.the_return' does not depend on any axioms -/
#guard_msgs in #print axioms TOE0.the_return
/-- info: 'TOE0.units_square_to_minus_one' does not depend on any
  axioms -/
#guard_msgs in #print axioms TOE0.units_square_to_minus_one
/-- info: 'TOE0.conj_involutive_on_the_chart' does not depend on any
  axioms -/
#guard_msgs in #print axioms TOE0.conj_involutive_on_the_chart
/-- info: 'TOE0.the_line_is_the_fixed_set' does not depend on any
  axioms -/
#guard_msgs in #print axioms TOE0.the_line_is_the_fixed_set
/-- info: 'TOE0.twenty_four_units_twelve_gates' does not depend on
  any axioms -/
#guard_msgs in #print axioms TOE0.twenty_four_units_twelve_gates
/-- info: 'TOE0.the_shell_splits_twelve_and_twelve' does not depend
  on any axioms -/
#guard_msgs in #print axioms TOE0.the_shell_splits_twelve_and_twelve
/-- info: 'TOE0.the_class_equation' does not depend on any axioms -/
#guard_msgs in #print axioms TOE0.the_class_equation
/-- info: 'TOE0.no_level_above' does not depend on any axioms -/
#guard_msgs in #print axioms TOE0.no_level_above
/-- info: 'TOE0.the_master_seal' does not depend on any axioms -/
#guard_msgs in #print axioms TOE0.the_master_seal

```

The root's axes are proved in the second kernel, R.2, *Triaxial Actuation.lean* (32 theorems, every one on no axiom; SHA-256 566615a6b9610fc981e03eb42e07af9e03849f20704729880f2bfe6434f2ea0b). For every actuation, the direction of its deed gives the fold, the registration and the seat; the registration is the only height-keeping map onto the seat; the seat is the only place both blind spots cancel; no reading of the record returns the orientation; and every self-grounding root is the root of an actuation over its own deeds, so the root carries the three axes and is not characterless (actuation_is_triaxial, registration_is_forced, the_only_special_cut, tongue_freedom, every_root_actuates, the_root_carries_three_axes). The root's grade is read in the third kernel, R.3, *Root Grade Ledger.lean* (18 theorems, every one on no axiom; SHA-256 c18b68b29d2b6be5058d56d3a7f9e966bcde6ce099217e0f5945db4c829c42b8). On the grade ledger of the Master Codex (Islam 2026n), carried with its weakest-link join law, the Codex's rule reads a

warranted root, one with a proof of its statement, at theorem grade, and an unwarranted root, the Codex's own declared axiom, at premise grade; the root of R.1 is warranted by a theorem with no axiom and no hypothesis, so on that rule it stands at theorem grade, the grade is read from the evidence the root is entered with, a warrant reading theorem and a posit premise, and not elected, and RA-Tongue co-location inherits the root's grade exactly (a_warranted_root_is_theorem_grade, the_ledger_reads_the_warrant, the_grade_is_not_elected, ra_is_theorem_grade, coloc_grade_is_the_roots_grade, the_ledger_seal). Both kernels compile with exit 0 and no message on Lean 4.19.0 and on Lean 4.22.0, declare no axiom, import nothing, and pin every cone as *does not depend on any axioms*; both are carried from the master plan of the series, Seven Rows, One Root (Islam 2026m), each extended in this revision of the series by one additive section, R.2 by its section IX and R.3 by its section VII, carried alike by every paper of the series, and the paper that reads them on its row cites them for exactly what they state.

```

/-
Triaxial_Actuation.lean · actuation is triaxial

Core Lean 4: no import, no axiom declared, no sorry, no
native_decide, and every theorem rests on no
axiom at all, not even propositional extensionality, quotients or
choice.

An actuation is a self-grounding root whose acts carry a
direction: each act has a side and a height,
and reversing the deed reverses the side and keeps the height.
From the direction alone the kernel
constructs three axes and proves them forced:
I The fold: reversing the deed is an involution.
II The registration: it keeps the height and forgets the side;
it is blind to the fold, closed with
it under combination, and it is the only height-keeping map
onto the seat.
III The seat: the fold's fixed set, where the fold and the
registration both fix the act, and off
which neither does.
IV The third axis: off the seat the fold's orbit has two acts
with one record, so the record leaves
exactly the orientation open; every reading blind to the fold
reads both alike.
V The root through the axes: a reversed deed is still an act
and instances the root; the axes are
carried by every actuation, and they force no orientation.
VI The bridge atom: the integers, with negation as the fold, are
an actuation.
VII The triaxial seal.
VIII The Tongue: no reading of the record returns the orientation;
the seat is the only special cut.
IX Every root actuates: a self-grounding root is the root of an
actuation over its own acts, so the
root carries the three axes; the registration is the fold's
record and lands on the seat, where
the orientation an act and its reversal share stays open; the
even pair leaves two acts, the
orientation locks one; and a degenerate direction, one side
only, is the characterless case.
-/

namespace TriAct

/-! ## The root, self-grounding -/

/-- A self-grounding root: acts occur, and every act instances the
root. -/
structure SelfGrounding (R : Prop) where
  Act      : Type

```

```

  anAct      : Act
  instances  : Act → R

/-! ## Actuation: a self-grounding root whose acts carry a direction
-/

/-- The direction of a deed: a side with a reversal, whose one fixed
side is the zero side. -/
structure Direction where
  Side      : Type
  rev       : Side → Side
  zero      : Side
  rev_rev   : ∀ s, rev (rev s) = s
  rev_zero  : rev zero = zero
  fix_zero  : ∀ s, rev s = s → s = zero

/-- An actuation: a self-grounding root whose acts are pairs of a
side and a height. -/
structure Actuation (R : Prop) where
  dir       : Direction
  Height    : Type
  anAct     : dir.Side × Height
  instances : dir.Side × Height → R

variable {R : Prop}

/-- Every actuation is a self-grounding root. -/
def Actuation.toSelfGrounding (A : Actuation R) : SelfGrounding R :=
  (A.dir.Side × A.Height, A.anAct, A.instances)

/-- The fold: reversing the deed reverses the side and keeps the
height. -/
def fold (A : Actuation R) (p : A.dir.Side × A.Height) : A.dir.Side
× A.Height := (A.dir.rev p.1, p.2)

/-- The registration: it keeps the height and forgets the side. -/
def reg (A : Actuation R) (p : A.dir.Side × A.Height) : A.dir.Side ×
A.Height := (A.dir.zero, p.2)

/-- The seat: the acts the fold fixes. -/
def onSeat (A : Actuation R) (p : A.dir.Side × A.Height) : Prop :=
  fold A p = p

/-! ## I · the fold -/

theorem fold_involutive (A : Actuation R) (p : A.dir.Side ×
A.Height) : fold A (fold A p) = p := by
  cases p with
  | mk s h => show (A.dir.rev (A.dir.rev s), h) = (s, h); rw
[A.dir.rev_rev]

/-! ## II · the registration -/

/-- The registration forgets the side: an act and its reversal have
one record. -/
theorem reg_forgets_side (A : Actuation R) (p : A.dir.Side ×
A.Height) : reg A (fold A p) = reg A p := rfl

/-- The registration keeps the height. -/
theorem reg_keeps_height (A : Actuation R) (p : A.dir.Side ×
A.Height) : (reg A p).2 = p.2 := rfl

/-- The even pair is closed under combination: folding a record
changes nothing. -/
theorem fold_of_reg (A : Actuation R) (p : A.dir.Side × A.Height) :
fold A (reg A p) = reg A p := by
  show (A.dir.rev A.dir.zero, p.2) = (A.dir.zero, p.2); rw
[A.dir.rev_zero]

/-- The registration is idempotent. -/
theorem reg_idempotent (A : Actuation R) (p : A.dir.Side × A.Height) :
reg A (reg A p) = reg A p := rfl

/-- THE REGISTRATION IS FORCED, NOT ELECTED: any map that keeps the
height and lands on the seat is
the registration. -/
theorem registration_is_forced (A : Actuation R) (g : A.dir.Side ×
A.Height → A.dir.Side × A.Height)
  (keeps : ∀ p, (g p).2 = p.2) (lands : ∀ p, onSeat A (g p)) : ∀ p,
g p = reg A p := by
  intro p
  have hs : (g p).1 = A.dir.zero := by
    have hf := lands p
    have h1 : A.dir.rev (g p).1 = (g p).1 := congrArg Prod.fst hf

```

```

    exact A.dir.fix_zero _ h1
have hh := keeps p
cases hg : g p with
| mk s h =>
  rw [hg] at hs hh
  show (s, h) = (A.dir.zero, p.2)
  rw [show s = A.dir.zero from hs, show h = p.2 from hh]

/-! ## III · the seat -/

/-- The seat is exactly the zero side. -/
theorem seat_iff_zero_side (A : Actuation R) (p : A.dir.Side ×
  A.Height) : OnSeat A p ↔ p.1 = A.dir.zero :=
  (fun h => A.dir.fix_zero _ (congrArg Prod.fst h),
  fun h => by
    cases p with
    | mk s t =>
      show (A.dir.rev s, t) = (s, t)
      have hs : s = A.dir.zero := h
      rw [hs, A.dir.rev_zero])

/-- ON THE SEAT BOTH BLIND SPOTS CANCEL: the fold and the
  registration both fix the act. -/
theorem seat_cancels_both (A : Actuation R) (p : A.dir.Side ×
  A.Height) (h : OnSeat A p) :
  fold A p = p ∧ reg A p = p := by
  refine (h, ?_)
  have hz : p.1 = A.dir.zero := (seat_iff_zero_side A p).mp h
  cases p with
  | mk s t =>
    have hs : s = A.dir.zero := hz
    show (A.dir.zero, t) = (s, t)
    rw [hs]

/-- OFF THE SEAT BOTH STAND: neither the fold nor the registration
  fixes the act. -/
theorem off_seat_both (A : Actuation R) (p : A.dir.Side × A.Height)
  (h : ¬ OnSeat A p) :
  fold A p ≠ p ∧ reg A p ≠ p := by
  refine (h, fun hr => h ?_)
  have hz : p.1 = A.dir.zero := (congrArg Prod.fst hr).symm
  exact (seat_iff_zero_side A p).mpr hz

/-! ## IV · the third axis -/

/-- THE RECORD LEAVES EXACTLY THE ORIENTATION: off the seat, an act
  and its reversal are two acts with
  one record. -/
theorem record_leaves_the_orientation (A : Actuation R) (p :
  A.dir.Side × A.Height) (h : ¬ OnSeat A p) :
  fold A p ≠ p ∧ reg A (fold A p) = reg A p :=
  (h, rfl)

/-- Every reading blind to the fold reads an act and its reversal
  alike. -/
theorem even_readings_are_blind (A : Actuation R) {α : Type} (f :
  A.dir.Side × A.Height → α)
  (blind : ∀ p, f (fold A p) = f p) (p : A.dir.Side × A.Height) :
  f (fold A p) = f p :=
  blind p

/-- The registration is such a reading, and so is every reading of
  the record. -/
theorem readings_of_the_record_are_blind (A : Actuation R) {α :
  Type} (k : A.dir.Side × A.Height → α)
  (p : A.dir.Side × A.Height) : k (reg A (fold A p)) = k (reg A p)
  := rfl

/-! ## V · the root through the axes -/

/-- A REVERSED DEED IS STILL AN ACT, AND INSTANCES THE ROOT: the
  denial spends the deed. -/
theorem denial_instances_root (A : Actuation R) (p : A.dir.Side ×
  A.Height) : R :=
  A.instances (fold A p)

/-- The root holds on every actuation, by its act. -/
theorem root_holds (A : Actuation R) : R := A.instances A.anAct

/-- THE AXES FORCE NO ORIENTATION: off the seat both orientations
  are acts, both instance the root, and
  the record reads them alike; the triaxial structure leaves the one
  orientation open. -/

```

```

theorem axes_force_no_orientation (A : Actuation R) (p : A.dir.Side
  × A.Height) (h : ¬ OnSeat A p) :
  R ∧ R ∧ fold A p ≠ p ∧ reg A (fold A p) = reg A p :=
  (A.instances p, A.instances (fold A p), h, rfl)

/-! ## VI · the bridge atom: the integers are an actuation -/

theorem int_neg_neg : ∀ s : Int, -(-s) = s
| Int.ofNat 0 => rfl
| Int.ofNat (_ + 1) => rfl
| Int.negSucc _ => rfl

theorem self_neg_zero : ∀ v : Int, -v = v → v = 0
| Int.ofNat 0, _ => rfl
| Int.ofNat (_ + 1), h => Int.noConfusion h
| Int.negSucc _, h => Int.noConfusion h

/-- The integer direction: negation, with zero its one fixed side. -/
def intDir : Direction := (Int, fun s => -s, 0, int_neg_neg, rfl,
  self_neg_zero)

/-- The bridge atom: points (d, t), the fold (d, t) ↔ (-d, t), the
  registration (d, t) ↔ (0, t). -/
def atom : Actuation True := (intDir, Int, ((0 : Int), (0 : Int)),
  fun _ => True.intro)

/-- THE BRIDGE ATOM, AS AN ACTUATION: the fold fixes (d, t) exactly
  when d = 0, the registration
  leaves it unchanged exactly then, and off the seat the record leaves
  the orientation. -/
theorem the_bridge_atom (d t : Int) :
  (OnSeat atom (d, t) ↔ d = 0) ∧ (reg atom (d, t) = (d, t) ↔ d = 0)
  ∧
  (d ≠ 0 → fold atom (d, t) ≠ (d, t) ∧ reg atom (fold atom (d, t))
  = reg atom (d, t)) :=
  (seat_iff_zero_side atom (d, t),
  (fun h => (congrArg Prod.fst h).symm, fun h => by rw [h]; rfl),
  fun hd => record_leaves_the_orientation atom (d, t) (fun hs => hd
  ((seat_iff_zero_side atom (d, t)).mp hs)))

/-- The atom is not characterless: off the seat it has two acts with
  one record. -/
theorem the_atom_is_not_characterless :
  fold atom ((1 : Int), (0 : Int)) ≠ ((1 : Int), (0 : Int)) ∧ reg
  atom (fold atom ((1 : Int), (0 : Int))) = reg atom ((1 : Int), (0
  : Int)) :=
  (fun h => Int.noConfusion (congrArg Prod.fst h), rfl)

/-! ## VII · the triaxial seal -/

/-- ACTUATION IS TRIAXIAL: every actuation carries a fold that is an
  involution, a registration that
  keeps the height, forgets the side and is the only such map onto the
  seat, and a seat where both fix
  the act and off which neither does; off the seat the record leaves
  exactly the orientation; every act
  and its reversal instance the root; and the axes force no
  orientation. -/
theorem actuation_is_triaxial (A : Actuation R) :
  (∀ p, fold A (fold A p) = p) ∧
  (∀ p, reg A (fold A p) = reg A p ∧ (reg A p).2 = p.2 ∧ fold A
  (reg A p) = reg A p) ∧
  (∀ g : A.dir.Side × A.Height → A.dir.Side × A.Height,
  (∀ p, (g p).2 = p.2) → (∀ p, OnSeat A (g p) → ∀ p, g p = reg
  A p) ∧
  (∀ p, OnSeat A p → fold A p = p ∧ reg A p = p) ∧
  (∀ p, ¬ OnSeat A p → fold A p ≠ p ∧ reg A p ≠ p) ∧
  (∀ p, ¬ OnSeat A p → R ∧ R ∧ fold A p ≠ p ∧ reg A (fold A p) =
  reg A p) :=
  (fold_involutive A, fun p => (reg_forgets_side A p,
  reg_keeps_height A p, fold_of_reg A p),
  registration_is_forced A, seat_cancels_both A, off_seat_both A,
  axes_force_no_orientation A)

/-- THE TRIAXIAL SEAL: actuation is triaxial, universally; the
  bridge atom is an actuation; and it is
  not characterless. -/
theorem the_triaxial_seal :
  (∀ (R : Prop) (A : Actuation R), (∀ p, fold A (fold A p) = p) ∧
  (∀ p, OnSeat A p → fold A p = p ∧ reg A p = p) ∧
  (∀ p, ¬ OnSeat A p → fold A p ≠ p ∧ reg A (fold A p) = reg A
  p)) ∧
  (∀ d t : Int, OnSeat atom (d, t) ↔ d = 0) ∧

```

```

(fold atom ((1 : Int), (0 : Int)) ≠ ((1 : Int), (0 : Int)) ∧ reg
atom (fold atom ((1 : Int), (0 : Int))) = reg atom ((1 : Int), (0
: Int))) :=
(fun _ A => (fold_involutive A, seat_cancels_both A,
fun p h => record_leaves_the_orientation A p h),
fun d t => seat_iff_zero_side atom (d, t),
the_atom_is_not_characterless)

/-! ## VIII · the Tongue: freedom and the only special cut -/

/-- TONGUE FREEDOM: no reading of the record returns the
orientation. If a record is blind to the fold
and an orientation is flipped by it, the orientation is not a
function of the record. -/
theorem tongue_freedom {α β : Type} (τ : α → α) (ρ : α → β) (g : β →
Bool) (d : α → Bool) (x : α)
| blind : ρ (τ x) = ρ x (f!ip : d (τ x) = !d x) : d ≠ fun y => g
(p y) := by
intro h
have h1 : d (τ x) = g (ρ (τ x)) := congrFun h (τ x)
have h2 : d x = g (ρ x) := congrFun h x
rw [blind] at h1
rw [← h2] at h1
rw [h1] at f!ip
cases hd : d x with
| false => rw [hd] at f!ip; exact Bool.noConfusion f!ip
| true => rw [hd] at f!ip; exact Bool.noConfusion f!ip

/-- THE ONLY SPECIAL CUT: the seat is exactly where both blind spots
cancel; nowhere else do the fold
and the registration both fix an act. -/
theorem the_only_special_cut (A : Actuation R) (p : A.dir.Side ×
A.Height) :
(fold A p = p ∧ reg A p = p) ↔ OnSeat A p :=
(fun h => h.1, fun h => seat_cancels_both A p h)

/-! ## IX · every root actuates: the root carries the three axes -/

/-- THE REGISTRATION IS THE FOLD'S RECORD: it reads an act and its
reversal alike and keeps the height, and two
acts have one record exactly when they have one height. -/
theorem registration_is_the_folds_record (A : Actuation R) (p q :
A.dir.Side × A.Height) :
reg A (fold A p) = reg A p ∧ (reg A p = reg A q ↔ p.2 = q.2) :=
(rfl, (fun h => show (reg A p).2 = (reg A q).2 from congrArg
Prod.snd h, fun h => by
cases p with
| mk s t =>
cases q with
| mk s' t' =>
have ht : t = t' := h
show (A.dir.zero, t) = (A.dir.zero, t')
rw [ht]))

/-- THE REGISTRATION LANDS ON THE SEAT: every record is a seat
point. -/
theorem reg_lands_on_seat (A : Actuation R) (p : A.dir.Side ×
A.Height) : OnSeat A (reg A p) :=
fold_of_reg A p

/-- THE ORIENTATION IS OPEN AT THE SEAT: off the seat, an act and
its reversal land on one seat point, the record
of both; at that place the record has left exactly the orientation
open. -/
theorem open_at_the_seat (A : Actuation R) (p : A.dir.Side ×
A.Height) (h : ¬ OnSeat A p) :
OnSeat A (reg A p) ∧ reg A (fold A p) = reg A p ∧ fold A p ≠ p :=
(reg_lands_on_seat A p, rfl, h)

/-- TWO AXES LEAVE TWO: off the seat, every reading blind to the
fold, the record among them, reads an act and
its reversal alike: two acts, one reading. -/
theorem even_pair_leaves_two (A : Actuation R) (p : A.dir.Side ×
A.Height) (h : ¬ OnSeat A p) {α : Type}
(f : A.dir.Side × A.Height → α) (blind : ∀ q, f (fold A q) = f
q) :
fold A p ≠ p ∧ f (fold A p) = f p ∧ reg A (fold A p) = reg A p :=
(h, blind p, rfl)

/-- THREE AXES LOCK ONE: the record together with the orientation,
the side, determines the act. -/
theorem orientation_locks_the_act (A : Actuation R) (p q :
A.dir.Side × A.Height)

```

```

(hr : reg A p = reg A q) (hs : p.1 = q.1) : p = q := by
cases p with
| mk s t =>
cases q with
| mk s' t' =>
have ht : t = t' := congrArg Prod.snd hr
have hs' : s = s' := hs
rw [hs', ht]

/-- A degenerate direction has the zero side and no other. -/
def Direction.Degenerate (D : Direction) : Prop := ∀ s, s = D.zero

/-- THE CHARACTERLESS CASE: over a degenerate direction every act is
on the seat, and the fold and the
registration are both the identity: one scalar, no fold, no seat to
lock. -/
theorem degenerate_is_characterless (A : Actuation R) (h :
A.dir.Degenerate) (p : A.dir.Side × A.Height) :
OnSeat A p ∧ fold A p = p ∧ reg A p = p :=
have hs : OnSeat A p := (seat_iff_zero_side A p).mpr (h p.1)
(hs, seat_cancels_both A p hs)

/-- The integer direction is not degenerate. -/
theorem intDir_not_degenerate : ¬ intDir.Degenerate := fun h =>
have h0 : (1 : Int) = 0 := h (1 : Int)
absurd h0 (by decide)

/-- EVERY ROOT ACTUATES: a self-grounding root is the root of the
actuation over its own acts, with the integer
direction; the root's deeds sit at the zero side and instance the
root as before. -/
def SelfGrounding.toActuation (S : SelfGrounding R) : Actuation R :=
(intDir, S.Act, ((0 : Int), S.anAct), fun p => S.instances p.2)

theorem every_root_actuates (S : SelfGrounding R) :
(∀ a : S.Act, OnSeat S.toActuation ((0 : Int), a)) ∧
(∀ a : S.Act, ¬ OnSeat S.toActuation ((1 : Int), a)) ∧
¬ S.toActuation.dir.Degenerate :=
(fun _ => rfl,
fun a h =>
have h0 : (1 : Int) = 0 := (seat_iff_zero_side S.toActuation
((1 : Int), a)).mp h
absurd h0 (by decide),
intDir_not_degenerate)

/-- THE ROOT CARRIES THREE AXES: for every self-grounding root, its
actuation carries the fold, an involution; the
registration, keeping the height, forgetting the side, the only
height-keeping map onto the seat; and the seat,
where both fix the act; and it is not characterless: at every height
an act off the seat, its reversal a second
act with the same record, both instanting the root, the orientation
between them forced by nothing. -/
theorem the_root_carries_three_axes (S : SelfGrounding R) :
(∀ p, fold S.toActuation (fold S.toActuation p) = p) ∧
(∀ p, reg S.toActuation (fold S.toActuation p) = reg
S.toActuation p ∧ (reg S.toActuation p).2 = p.2) ∧
(∀ g : Int × S.Act → Int × S.Act, (∀ p, (g p).2 = p.2) → (∀ p,
OnSeat S.toActuation (g p)) →
∀ p, g p = reg S.toActuation p) ∧
(∀ p, OnSeat S.toActuation p → fold S.toActuation p = p ∧ reg
S.toActuation p = p) ∧
(∀ a : S.Act, fold S.toActuation ((1 : Int), a) ≠ ((1 : Int), a) ∧
reg S.toActuation (fold S.toActuation ((1 : Int), a)) = reg
S.toActuation ((1 : Int), a)) ∧
(∀ p, ¬ OnSeat S.toActuation p →
R ∧ R ∧ fold S.toActuation p ≠ p ∧ reg S.toActuation (fold
S.toActuation p) = reg S.toActuation p) :=
(fold_involutive S.toActuation,
fun p => (reg_forgets_side S.toActuation p, reg_keeps_height
S.toActuation p),
registration_is_forced S.toActuation,
seat_cancels_both S.toActuation,
fun a => record_leaves_the_orientation S.toActuation ((1 : Int),
a) ((every_root_actuates S).2.1 a),
axes_force_no_orientation S.toActuation)

end TriAct

/-! ## Cones, pinned as printed: every theorem on no axiom -/
/-- info: 'TriAct.fold_involutive' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.fold_involutive
/-- info: 'TriAct.reg_forgets_side' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.reg_forgets_side

```

```

/-- info: 'TriAct.reg_keeps_height' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.reg_keeps_height
/-- info: 'TriAct.fold_of_reg' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.fold_of_reg
/-- info: 'TriAct.reg_idempotent' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.reg_idempotent
/-- info: 'TriAct.registration_is_forced' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.registration_is_forced
/-- info: 'TriAct.seat_iff_zero_side' does not depend on any axioms
-/
#guard_msgs in #print axioms TriAct.seat_iff_zero_side
/-- info: 'TriAct.seat_cancels_both' does not depend on any axioms
-/
#guard_msgs in #print axioms TriAct.seat_cancels_both
/-- info: 'TriAct.off_seat_both' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.off_seat_both
/-- info: 'TriAct.record_leaves_the_orientation' does not depend on
any axioms -/
#guard_msgs in #print axioms TriAct.record_leaves_the_orientation
/-- info: 'TriAct.even_readings_are_blind' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.even_readings_are_blind
/-- info: 'TriAct.readings_of_the_record_are_blind' does not depend
on any axioms -/
#guard_msgs in #print axioms TriAct.readings_of_the_record_are_blind
/-- info: 'TriAct.denial_instances_root' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.denial_instances_root
/-- info: 'TriAct.root_holds' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.root_holds
/-- info: 'TriAct.axes_force_no_orientation' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.axes_force_no_orientation
/-- info: 'TriAct.int_neg_neg' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.int_neg_neg
/-- info: 'TriAct.self_neg_zero' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.self_neg_zero
/-- info: 'TriAct.the_bridge_atom' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.the_bridge_atom
/-- info: 'TriAct.the_atom_is_not_characterless' does not depend on
any axioms -/
#guard_msgs in #print axioms TriAct.the_atom_is_not_characterless
/-- info: 'TriAct.actuation_is_tri axial' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.actuation_is_tri axial
/-- info: 'TriAct.the_tri axial_seal' does not depend on any axioms
-/
#guard_msgs in #print axioms TriAct.the_tri axial_seal
/-- info: 'TriAct.tongue_freedom' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.tongue_freedom
/-- info: 'TriAct.the_only_special_cut' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.the_only_special_cut
/-- info: 'TriAct.registration_is_the_folds_record' does not depend
on any axioms -/
#guard_msgs in #print axioms TriAct.registration_is_the_folds_record
/-- info: 'TriAct.reg_lands_on_seat' does not depend on any axioms
-/
#guard_msgs in #print axioms TriAct.reg_lands_on_seat
/-- info: 'TriAct.open_at_the_seat' does not depend on any axioms -/
#guard_msgs in #print axioms TriAct.open_at_the_seat
/-- info: 'TriAct.even_pair_leaves_two' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.even_pair_leaves_two
/-- info: 'TriAct.orientation_locks_the_act' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.orientation_locks_the_act
/-- info: 'TriAct.degenerate_is_characterless' does not depend on
any axioms -/
#guard_msgs in #print axioms TriAct.degenerate_is_characterless
/-- info: 'TriAct.intDir_not_degenerate' does not depend on any
axioms -/
#guard_msgs in #print axioms TriAct.intDir_not_degenerate
/-- info: 'TriAct.every_root_actuates' does not depend on any axioms
-/
#guard_msgs in #print axioms TriAct.every_root_actuates
/-- info: 'TriAct.the_root_carries_three_axes' does not depend on
any axioms -/
#guard_msgs in #print axioms TriAct.the_root_carries_three_axes
-/
Root_Grade_Ledger.lean · the root at theorem grade, and RA-Tongue
co-location with it

```

Core Lean 4: no import, no axiom declared, no sorry, no native_decide, and every theorem rests on no axiom at all, not even propositional extensionality, quotients or choice.

The floor Codex (Master Codex v5.1.0, Codex.lean) grades every claim on one ledger and joins claims at the weakest link. It declares the root as 'axiom RA' and pins 'raClaim.grade = .premise'; its co-location claim, the root joined with the wall, therefore inherits '.premise'. This kernel carries the same ledger and the same join law with no axiom declared, warrants the root by a theorem with no axiom and no hypothesis, and proves that the co-location claim then stands at theorem grade.

- I The grade ledger and the weakest-link join, as in the Codex.
- II The root, a theorem: the constructed domain of TOE_Zero.lean, warranted on no axiom.
- III The wall, a theorem: the Codex's K4 frame, its target not a function of its record, proved on no axiom through Tongue freedom.
- IV The claims: the root at theorem grade, the wall at theorem grade, and their co-location.
- V Why the Codex read premise: a join's grade is its weakest link, and co-location's grade is the root's grade exactly; with the root at theorem grade, co-location is a theorem.
- VI The ledger seal.
- VII The grade read from the warrant, as the Codex reads it: a root with a warrant, a proof of its statement, is theorem grade; a root posited without one, the Codex's 'axiom RA', is premise grade; the two claims of section IV carry exactly the grades this rule reads.

-/

namespace RGL

/-! ## I · the grade ledger, as in the Codex -/

inductive Grade : Type
| premise | corroboration | operational | structural | engineering
| conditional | theoremConditional | analytic | theorem
deriving DecidableEq

```

def Grade.rank : Grade → Nat
| .premise => 0 | .corroboration => 1 | .operational => 2 |
|.structural => 3
| .engineering => 4 | .conditional => 5 | .theoremConditional => 6
| .analytic => 7 | .theorem => 8

```

```

def Grade.weakest (a b : Grade) : Grade := if a.rank ≤ b.rank then a
else b

```

/- A claim carries its statement, its grade and its warrant, a proof of the statement. -/

```

structure Claim : Type where
stmt    : Prop
grade   : Grade
warrant : stmt

```

/- Two claims join at their weakest link. -/

```

def Claim.join (c₁ c₂ : Claim) : Claim where
stmt    := c₁.stmt ∧ c₂.stmt
grade   := Grade.weakest c₁.grade c₂.grade
warrant := (c₁.warrant, c₂.warrant)

```

```

theorem join_grade (c₁ c₂ : Claim) : (Claim.join c₁ c₂).grade =
Grade.weakest c₁.grade c₂.grade := rfl

```

/-! ## II · the root, a theorem (the constructed domain of TOE_Zero.lean) -/

```

structure SelfGrounding (R : Prop) where
Act      : Type
anAct    : Act
instances : Act → R

```

```

theorem seated_undeniable (R : Prop) (G : SelfGrounding R) : R :=
G.instances G.anAct

```

```

def ΔE₀ : Unit → Int := fun _ => 1
def RA₀ : Prop := ∀ x : Unit, 0 < ΔE₀ x

```

```

def ra0 : SelfGrounding RA0 := (Unit, ()), fun _ => show (0 : Int)
  < 1 by decide)

/-- THE ROOT ON THE CONSTRUCTED DOMAIN, A THEOREM: no axiom, no
hypothesis. -/
theorem root_on_the_constructed_domain : RA0 := seated_undeniable
  ra0

/-! ## III · the wall, a theorem (the Codex's K4 frame) -/

/-- Tongue freedom: if a record is blind to the fold and an
orientation is flipped by it, the
orientation is not a function of the record. -/
theorem tongue_freedom {α β : Type} (τ : α → α) (ρ : α → β) (g : β →
  Bool) (d : α → Bool) (x : α)
  (blind : ρ (τ x) = ρ x) (flip : d (τ x) = !d x) : d ≠ fun y => g
  (ρ y) := by
  intro h
  have h1 : d (τ x) = g (ρ (τ x)) := congrFun h (τ x)
  have h2 : d x = g (ρ x) := congrFun h x
  rw [blind] at h1
  rw [h2] at h1
  rw [h1] at flip
  cases hd : d x with
  | false => rw [hd] at flip; exact Bool.noConfusion flip
  | true => rw [hd] at flip; exact Bool.noConfusion flip

namespace K4

def val : Nat → Nat
  | 0 => 11 | 1 => 851 | 2 => 13 | 3 => 1273 | 4 => 17 | 5 => 437
  | 6 => 19 | 7 => 2119 | 8 => 23 | 9 => 1703 | 10 => 29 | 11 => 869
  | _ => 0

def tau : Nat → Nat := fun i =>
  if i < 12 then (if i % 2 = 0 then i + 1 else i - 1) else i

def target : Nat → Bool := fun i => i % 2 == 0

def rho : Nat → Nat := fun i =>
  (val i % 2) * 1000 + (val i % 3) * 100 + (val i % 5) * 10 + (val i
  % 7)

/-- The record is blind to the fold at the first pair. -/
theorem register_even_at_zero : rho (tau 0) = rho 0 := by decide

/-- The target is flipped by the fold at the first pair. -/
theorem target_odd_at_zero : target (tau 0) = !target 0 := by decide

/-- THE WALL: no reading of the record returns the target. -/
theorem wall (g : Nat → Bool) : target ≠ fun y => g (rho y) :=
  tongue_freedom tau rho g target 0 register_even_at_zero
  target_odd_at_zero

end K4

/-! ## IV · the claims -/

/-- The root at theorem grade, warranted by a theorem with no axiom
and no hypothesis. -/
def raClaim0 : Claim where
  stmt := RA0
  grade := .theorem
  warrant := root_on_the_constructed_domain

/-- The wall at theorem grade, as in the Codex. -/
def wallClaim : Claim where
  stmt := ∀ g : Nat → Bool, K4.target ≠ fun y => g (K4.rho y)
  grade := .theorem
  warrant := K4.wall

/-- RA-Tongue co-location: the root joined with the wall, as in the
Codex. -/
def colocClaim0 : Claim := Claim.join raClaim0 wallClaim

/-- The Codex's root, graded as the Codex grades it, for comparison:
the same statement form, at
premise grade. -/
def raClaimPremise : Claim where
  stmt := RA0
  grade := .premise
  warrant := root_on_the_constructed_domain

/-! ## V · why the Codex read premise, and what co-location is now -/

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```

/-- A join with a theorem-grade claim has exactly the other claim's
grade. -/
theorem weakest_with_theorem (g : Grade) : Grade.weakest g .theorem
  = g := by
  cases g <;> rfl

/-- CO-LOCATION'S GRADE IS THE ROOT'S GRADE, EXACTLY: the wall is a
theorem, so the join reads the
root's grade and nothing else. -/
theorem coloc_grade_is_the_roots_grade (r : Claim) :
  (Claim.join r wallClaim).grade = r.grade := by
  show Grade.weakest r.grade Grade.theorem = r.grade
  exact weakest_with_theorem r.grade

/-- WHY THE CODEX READ PREMISE: with the root labelled premise, the
co-location claim is premise. -/
theorem premise_root_gives_premise_coloc : (Claim.join
  raClaimPremise wallClaim).grade = .premise := rfl

/-- THE ROOT IS A THEOREM-GRADE CLAIM. -/
theorem ra_is_theorem_grade : raClaim0.grade = .theorem := rfl

/-- RA-TONGUE CO-LOCATION IS A THEOREM-GRADE CLAIM. -/
theorem coloc_is_theorem_grade : colocClaim0.grade = .theorem := rfl

/-- The co-location claim's statement holds: the root and the wall,
both proved. -/
theorem coloc_holds : colocClaim0.stmt := colocClaim0.warrant

/-! ## VI · the ledger seal -/

/-- THE LEDGER SEAL: the root is a theorem on no axiom and no
hypothesis; the wall is a theorem; their
co-location holds and stands at theorem grade; and co-location's
grade is the root's grade exactly,
which is why the Codex, labelling the root premise, read premise. -/
theorem the_ledger_seal :
  RA0 ∧ (∀ g : Nat → Bool, K4.target ≠ fun y => g (K4.rho y)) ∧
  colocClaim0.stmt ∧ colocClaim0.grade = .theorem ∧
  (∀ r : Claim, (Claim.join r wallClaim).grade = r.grade) ∧
  (Claim.join raClaimPremise wallClaim).grade = .premise :=
  (root_on_the_constructed_domain, K4.wall, coloc_holds,
  coloc_is_theorem_grade,
  coloc_grade_is_the_roots_grade, premise_root_gives_premise_coloc)

/-! ## VII · the grade read from the warrant, as the Codex reads it
-/

/-- The evidence a ledger entry carries for a statement: posited
with no warrant, or warranted by a proof. -/
inductive Evidence (P : Prop) : Type
  | posited
  | warranted (w : P)

/-- THE CODEX'S RULE FOR THE ROOT'S GRADE: evidence at theorem grade
is a proof with its axioms printed. A root
that comes with a warrant, a proof of its statement, is theorem
grade; a root posited with none, as the Codex
declares "axiom RA", is premise grade. -/
def rootGrade : Evidence RA0 → Grade
  | .warranted _ => .theorem
  | .posited _ => .premise

/-- A WARRANTED ROOT IS THEOREM GRADE: the root of section II
carries its warrant, so the rule reads theorem. -/
theorem a_warranted_root_is_theorem_grade :
  rootGrade (.warranted root_on_the_constructed_domain) = .theorem
  := rfl

/-- AN UNWARRANTED ROOT IS PREMISE GRADE: with no warrant the rule
reads premise, the Codex's own reading of
its declared axiom. -/
theorem an_unwarranted_root_is_premise_grade : rootGrade .posited =
  .premise := rfl

/-- THE LEDGER READS THE WARRANT: the root claim of section IV
carries the grade the rule reads from its warrant,
and the comparison claim carries the grade the rule reads from none.
-/
theorem the_ledger_reads_the_warrant :
  raClaim0.grade = rootGrade (.warranted raClaim0.warrant) ∧
  raClaimPremise.grade = rootGrade .posited :=
  (rfl, rfl)

```

```

/-- THE GRADE IS NOT ELECTED: the rule reads theorem from a warrant
and premise from none, never the other way;
whatever label an entry carries, the warranted root's grade under
the rule is theorem. -/
theorem the_grade_is_not_elected :
  (∀ w : RA₀, rootGrade (.warranted w) = .theorem) ∧ rootGrade
    .posited ≠ .theorem ∧
  (∀ w : RA₀, rootGrade (.warranted w) ≠ rootGrade .posited) :=
  (fun _ => rfl, fun h => Grade.noConfusion h, fun _ h =>
    Grade.noConfusion h)

end RGL

/-! ## Cones, pinned as printed: every theorem on no axiom -/
/-- info: 'RGL.join_grade' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.join_grade
/-- info: 'RGL.seated_undeniable' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.seated_undeniable
/-- info: 'RGL.root_on_the_constructed_domain' does not depend on
any axioms -/
#guard_msgs in #print axioms RGL.root_on_the_constructed_domain
/-- info: 'RGL.tongue_freedom' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.tongue_freedom
/-- info: 'RGL.K4.register_even_at_zero' does not depend on any
axioms -/
#guard_msgs in #print axioms RGL.K4.register_even_at_zero
/-- info: 'RGL.K4.target_odd_at_zero' does not depend on any axioms
-/
#guard_msgs in #print axioms RGL.K4.target_odd_at_zero
/-- info: 'RGL.K4.wall' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.K4.wall
/-- info: 'RGL.weakest_with_theorem' does not depend on any axioms
-/
#guard_msgs in #print axioms RGL.weakest_with_theorem
/-- info: 'RGL.coloc_grade_is_the_roots_grade' does not depend on
any axioms -/
#guard_msgs in #print axioms RGL.coloc_grade_is_the_roots_grade
/-- info: 'RGL.premise_root_gives_premise_coloc' does not depend on
any axioms -/
#guard_msgs in #print axioms RGL.premise_root_gives_premise_coloc
/-- info: 'RGL.ra_is_theorem_grade' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.ra_is_theorem_grade
/-- info: 'RGL.coloc_is_theorem_grade' does not depend on any axioms
-/
#guard_msgs in #print axioms RGL.coloc_is_theorem_grade
/-- info: 'RGL.coloc_holds' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.coloc_holds
/-- info: 'RGL.the_ledger_seal' does not depend on any axioms -/
#guard_msgs in #print axioms RGL.the_ledger_seal
/-- info: 'RGL.a_warranted_root_is_theorem_grade' does not depend on
any axioms -/
#guard_msgs in #print axioms RGL.a_warranted_root_is_theorem_grade
/-- info: 'RGL.an_unwarranted_root_is_premise_grade' does not depend
on any axioms -/
#guard_msgs in #print axioms
  RGL.an_unwarranted_root_is_premise_grade
/-- info: 'RGL.the_ledger_reads_the_warrant' does not depend on any
axioms -/
#guard_msgs in #print axioms RGL.the_ledger_reads_the_warrant
/-- info: 'RGL.the_grade_is_not_elected' does not depend on any
axioms -/
#guard_msgs in #print axioms RGL.the_grade_is_not_elected

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AUTHOR'S PROVENANCE AND METHOD DISCLOSURE

Author and method. Mohammad F. Islam, PhD, independent researcher, sole author and sole authority for every claim and every error. The paper was drafted with Claude (Anthropic) as scribe on the author's direction; the model checked the receipts cited from each paper and argued against the author's positions on instruction; it holds no authorship. The channel grading in the Discussion follows the two channels of Nothing Escapes Least Erasure (Islam 2026j), including its statement of the act's grade; the ledger departs from Islam (2026j) on rows 1, 5, 6 and 8 by the act, each row's

closure paper governing its row whatever the edition dates.

The register. In the programme's vocabulary: the block and hypothesis bits at $[\{\iota\}, T]$ on each row's kernel; the resolution bit by the act at premise grade, one keyed field per row, on Poincaré discharged by Perelman's witness at the grade of its citation, the kernel constructing no witness; the physical-witness column at the grades printed in Table 1; $\Delta M = 0$ on every cited result. This mini paper authors no theorem of its own and adds no warrant to the papers it reports; the additive sections of R.2 (section IX, nine theorems and two definitions) and R.3 (section VII, four theorems, one inductive type and one definition) were authored for the series in this revision and are carried alike by every paper of the series in the same editions; the three kernels of its Appendix R are the ones every row paper carries in the editions of record dated 6 October 2026, digest for digest.

Audit. The row papers claim no audit result; the cycles run in this programme are recorded in the audit log beside them; no external compilation of any kernel is yet on record.

Nature of Root Axiom

The root is triaxial, and that is the force of the cut. A characterless existence has no triaxiality: one scalar, no fold, no seat, nothing to lock. The root, to exist is to actuate, carries three axes, and they are not chosen: for every actuation, the direction of its deed gives the fold, the registration and the seat, and the registration is the only height-keeping map onto the seat. The deed is the direction: a denial spends the deed and instances the root. The registration is the fold's record: it keeps the height and forgets the side, and the two even axes, symmetry and record, are closed under combination and blind. The seat is the fixed set, where the two blind spots cancel, and the third axis is the one orientation the even pair cannot supply. Two axes never determine the point; three do. The seven rows are one cut because each is an involution with a seat and one missing orientation. The root makes the third axis native as a place: the seat, and the one orientation still open there; the axes themselves force no orientation. Which way that orientation points is the value, and it is the root read on that seat. The root alone still forces no value.

Key to the statement, in the words of its receipts. The three axes are the two even axes, the fold's symmetry and the record, and the orientation, the side of a point of the actuation, which the even pair cannot supply; the fold, the registration and the seat are what the direction of the deed gives, and the seat is where the side is read. The point two axes never determine is a point off the seat: there the even pair does not tell a point from its reversal, and the record with the side locks the point; on the seat both blind spots cancel and nothing is open there to determine. A point of the actuation is a side and a height, and every point instances the root: the points are the actuation's deeds, and the deed is the direction in the sense that its side is its direction, reversed by the fold. The characterless case, one scalar, is a degenerate direction: the fold is the identity and the seat is total, which is to say no fold and no seat to lock. The axes are not chosen in the sense that, given the direction, the fold, the registration and the seat are forced, the registration the only height-keeping map onto the seat; the root's own actuation takes the integer direction, under which it is not characterless, while a degenerate direction over the same deeds would be (every_root_actuates, degenerate_is_characterless). The root read on that seat is the root read on

the row at the seat point where a point and its reversal land with one record; that the orientation between them is the row's value bit is the paper's structural reading, a second statement beside its kernel's theorems, and no kernel states a map between the two. Each row carries the actuation's seat as the root of an actuation over its own deeds; a row's own seat, where its kernel states one, is a further statement of that kernel, named in the last row of the table, and on P versus NP it is vacant. A deed of the root is an inhabitant of its type of acts, which the root kernel's own comments call an act; in the root's actuation those deeds sit at the zero side, on the seat, and the points off the seat are the sides the direction supplies; the act, unqualified, is existence read on each row, the field its closing theorem consumes.

Receipts, each cited for what its statement says; R.1, R.2 and R.3 are the three kernels of Appendix R, every theorem of each on no axiom; the three theorems of the Master Codex are cited, not carried.

Clause of the statement	Receipt
the root carries three axes, not chosen: every self-grounding root is the root of an actuation over its own deeds, and for every actuation, given the direction of its deed, the fold, the registration and the seat are forced, the registration the only height-keeping map onto the seat the registration is the only height-keeping map onto the seat the axes force no orientation	every_root_actuates, the_root_carries_three_axes, actuation_is_triaxial, registration_is_forced (Appendix R.2) registration_is_forced (R.2) axes_force_no_orientation (R.2)
the deed is the direction: a deed of the actuation is a point, a side and a height, its side its direction, reversed by the fold; a reversed point is still a deed and instances the root; a denial spends the deed and instances the root	fold, denial_instances_root (R.2); denial_reenacts_root, denial_instantiates (R.1)
the registration is the fold's record: it keeps the height, forgets the side and reads a point and its reversal alike; the fold is an involution; the even pair is closed under combination	registration_is_the_folds_record, reg_forgets_side, reg_keeps_height, fold_involution, fold_of_reg (R.2)
the two even axes are blind: no reading of the record returns the orientation; symmetry keyless, the line property keyed	readings_of_the_record_are_blind, tongue_freedom (R.2); symmetry_is_keyless, line_property_is_keyed, exactly_one_stands (Islam 2026n, Master Codex v5.1.0, commit 5e55fca, cited for their statements, not carried; that Codex declares its root as an axiom, this paper's kernels declare none)

Clause of the statement	Receipt
the seat is the fixed set, where both blind spots cancel; off the seat both stand	seat_iff_zero_side, seat_cancels_both, off_seat_both, the_only_special_cut (R.2)
the registration lands on the seat; off the seat a point and its reversal land on one seat point with one record, and the orientation is what stays open there; every reading of the record reads both alike	reg_lands_on_seat, open_at_the_seat, record_leaves_the_orientation, readings_of_the_record_are_blind (R.2)
the characterless case, no fold and no seat to lock: over a degenerate direction, one side only, the fold is the identity and every point is on the seat; the root's actuation is not degenerate	degenerate_is_characterless, intDir_not_degenerate, every_root_actuates (R.2)
the bridge atom is an actuation, and not characterless	the_bridge_atom, the_atom_is_not_characterless, the_triaxial_seal (R.2)
the scalar ground in the algebra, a third reading beside the actuation's: the Return, the line the conjugation fixes and the twelve gates, bound with the root the root at theorem grade on the Codex's ledger rule, the grade read from the evidence the root is entered with, a warrant reading theorem and a posit premise, and not elected, and RA--Tongue co-location inheriting it	the_return, the_line_is_the_fixed_set, twenty_four_units_twelve_gates, the_class_equation, the_master_seal (R.1) a_warranted_root_is_theorem_grade, the_ledger_reads_the_warrant, the_grade_is_not_elected, ra_is_theorem_grade, coloc_grade_is_the_roots_grade, the_ledger_seal (R.3)
two axes never determine the point; three do: off the seat the even pair does not tell a point from its reversal, and the record with the side, which carries the orientation, locks the point; on the seat both blind spots cancel and nothing is open to determine the root alone forces no value; the root read on each row is that row's value	even_pair_leaves_two, orientation_locks_the_act, seat_cancels_both (R.2); in their own coordinates, for independent axes in $\mathbb{F}_2^3$, two_axes_leave_two, three_axes_lock_one in the six row kernels and the P versus NP kernel, a second statement undeniable_root_forces_no_value in every row kernel; act_is_the_value in the six row kernels, root_at_zeros_is_least_erasure in the Riemann kernel, root_on_computation_is_sep in the P versus NP kernel
the seven rows are one cut: each row's root is an actuation, so each carries an involution, the fold, with a seat and one missing orientation	every_root_actuates, the_root_carries_three_axes (R.2), stated for every self-grounding root and read at the constructed root ra_0 , or the Riemann kernel's rootSelfGrounding of every row kernel, whose structure repeats R.2's SelfGrounding field for field; each row's own seat as its paper states it, in the hypothesis column of Table 1: a line, a diagonal, a mirror's fixed point, a centre, a round point; none of its own on Navier--Stokes and Yang--Mills; vacant on P versus NP; the row ledger of Islam (2026j, Book III), cited

Clause of the statement	Receipt

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